BIOS 4803 /BIOL 8803 Origins of Biodiversity

Spring 2027

3 Credits

Instructors

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Prerequisites

Graduate standing in Biological Sciences or consent of instructor.

Advanced undergraduates with strong backgrounds in ecology, evolutionary biology, or genetics are welcome. Prerequisites for undergraduates are Evolution (BIOS 3600) **or** both of Genetics (BIOS 2600) & Ecology (BIOS 2300) **or** permission from the instructor(s).

Course Scope and Objectives

In 1959, the great ecologist G. Evelyn Hutchinson wrote, “Why are there so many kinds of animals?” Now, nearly 70 years later, that question remains as profound and unanswered as it was then.

In this course, you will explore the fundamental ecological and evolutionary processes that generate and maintain biological diversity across scales, from genes to ecosystems. Students will examine how species originate, interact, and adapt to their environments through a modern synthesis of evolutionary theory and ecological principles. The course emphasizes both conceptual frameworks and empirical research, providing graduate students and advanced undergraduates with fundamental tools to understand biodiversity patterns and processes. Through readings of classic and contemporary literature, discussions, and case studies, students will develop a sophisticated understanding of how both ecological and evolutionary processes jointly shape the living world.

Each class will revolve around a basic concept, from which we have identified a set of core concepts that we think all biology graduate students leaving with a PhD from our department should have a working understanding of. Each class will have required readings, a short lecture portion, and student activity (e.g., higher level journal discussion, or some other effective active learning technique that will engage students).

Critically, each class will end with what we *don't* know. Unresolved questions are as important as settled ones, and learning to identify the frontier is a core skill for any research biologist.

By the end of this course, you will be able to:

- Identify and evaluate the major ecological and evolutionary processes responsible for generating and maintaining biodiversity
- Read and critically analyze primary literature spanning ecology, evolutionary biology, biogeography, and behavior
- Develop and articulate your own position on active debates in the field, supported by evidence
- Connect short-term micro-scale processes to macro-scale patterns, and recognize where those connections remain contested

- Identify unresolved questions at the leading edge of contemporary ecological and evolutionary research, and evaluate what kinds of evidence would advance them

Course Structure

There will be two class sessions per week. Each week revolves around a central concept identified as something every graduating biologist in our department should understand. Sessions combine a lecture portion with an active student component, such as primary literature discussion, structured debate, or a problem-solving exercise. Required readings are listed in the course schedule below.

Graduate students will complete an additional component each week (see Assignments, below).

Semester Assignments

Participation and Discussion

Active engagement in discussion is the core of this course. You are expected to arrive having read the assigned papers and prepared to discuss them. Strong participation means asking probing questions, engaging with your classmates' arguments, and being willing to revise your thinking based on evidence and discussion. Most of the credit for participation is awarded for good-faith engagement, not for having the "correct" answer (which, in many cases, we won't know).

Weekly Reading Responses

Each student is expected to read **two papers** of the required reading prior to class. From these readings, before each class, submit a brief (1-2 paragraph) written response to the assigned readings via Canvas. These should identify the central question each paper addresses, what you found most interesting or most surprising, and at least one question you still have after reading. Responses are graded on completion and good-faith engagement. Late responses will not be accepted.

Class Quizzes

There will be a hand-written quiz at the start of each class pertaining to the pre-class readings. Each quiz will be approximately 5 questions long and contain a mixture of multiple choice and short-answer questions.

Graduate Student Assignment: Weekly Synthesis Presentation

Each week, one graduate student will present a short 5-8 minute Powerpoint presentation on a recently published paper on the topic of that week's classes that the student will find themselves. The 5-8 minute presentation will come after the general introduction delivered by Dr. Anderson/Dr. Stroud and will be graded on (a) placing the recent study into the historical context of the topic [why did the scientists do the study], (b) identifying the advance(s) that the recent study has made [what did they find], and (c) review what is still unanswered [what should they do next].

Midterm and Final Oral Exams

All students will complete a 10 minute midterm oral exam and a 20 minute final oral exam with Drs. Anderson and Stroud.

In the oral exams, students will deliver a short seminar-style presentation on an unresolved question that they have chosen. Presentations should end with an explicit identification of what remains unknown and what immediate research priorities the student thinks will be most productive going forward. There is no single correct answer to either question; we are asking for the students to make a reasonable case for their recommendations. Following the presentation, students will be asked discussion-prompting questions on topics that have been raised in class. We are looking for (1) reasonable familiarity with the material, (2) the ability to think critically about existing data, (3) the ability to suggest or identify competing hypotheses, and (4) the ability to distinguish between hypotheses and their predictions.

In the midterm exam, the presentation will be **5 minutes** followed by an additional **5 minutes** of question/discussion. In the final exam, the presentation will be **8-10 minutes** followed by **12-10 minutes** of question/discussion.

We recognize that the oral exam will be new to many students. The purpose of the midterm is to help students acclimate to the oral evaluation format when the stakes are low (5% of grade). Our intention is not to grill students or trip them up in any way; it is rather to invite students to demonstrate their grasp of the material and to think aloud (with us) about various aspects surrounding the origins of biodiversity.

Oral Exam Rubric:

Points for each category ranked 1-5, 5 being strongest

Presentation

Presentation is well-organized and clearly explained: __
Student has identified an unanswered question: __
Question is placed in an appropriate broader context: __
Alternative hypotheses articulated: __
Student can correctly distinguish question, hypotheses, and predictions: __
Student has provided logical suggestions for future research: __

Oral Examination

Student shows basic grasp of most core concepts covered in syllabus: __
Student can use this understanding as basis for a wide-ranging informal discussion __
Student shows a strong understanding of at least one topic covered in the course __

Undergraduate Student Grading Scheme

Class quizzes: 15%
Participation and Discussion: 20%
Weekly Reading Responses: 35%
Midterm Oral Exam: 5%
Final Oral Exam: 25%

Graduate Student Grading Scheme

Class quizzes: 15%
Participation and Discussion: 20%
Weekly Reading Responses: 25%
In-class Paper Presentation: 10%
Midterm Oral Exam: 5%
Final Oral Exam: 25%

Final letter grades: A = 100–89.5%, B = 89.4–79.5%, C = 79.4–69.5%, D = 69.4–59.5%, F = 59.4–0%

AI Policy

AI writing tools are not permitted. The key skills you will develop in this class are how to critically review, synthesize, and identify unresolved topics in the fields of ecology and evolution. AI will not help you develop those skills. Growing empirical work suggests that it actively hinders your ability to do so. We will be reading for evidence of genuine (human) thinking.

Attendance Policy

Class attendance is an essential component of this course, which is based on dialogue. Absence is excused for legitimate reason communicated in a timely manner (sickness, emergencies, etc.). Unexcused absences will affect your grade via missed quizzes and participation/discussion marks. Students should enter each class having read the material and being prepared to speak and listen and engage with their classmates.

Academic Integrity

Georgia Tech aims to cultivate a community based on trust, academic integrity, and honor. Students are expected to act according to the highest ethical standards. For details, see Georgia Tech's Academic Honor Code (policylibrary.gatech.edu/student-life/academic-honor-code).

Any student suspected of plagiarism, academic dishonesty (including violation of the AI policy) will be reported to the Office of Student Integrity. Any time information or ideas are drawn from a source, that source must be cited.

We must stress that all of science runs on the honor system. When you submit a work to a scientific journal for publication, it is taken on good faith that you conducted the work as claimed. For this reason, accusations against your academic integrity can result in lasting damage to your prospects in science. Don’t risk it.

Statement of Intent for Inclusivity

As members of the Georgia Tech community, we are committed to creating a learning environment in which all students feel safe and included. We welcome your feedback on what we can stop, start, and continue doing to make this a seminar where every participant feels valued and can contribute fully.

Learning Accommodations

If you require special accommodations, contact the Office of Disability Services at (404) 894-2563 or disabilityservices.gatech.edu to obtain an accommodations letter. Please also email Dr. Stroud as early as possible so we can discuss your needs.

Course Schedule

**This is a dynamic schedule and will be updated as the semester progresses.*

We will provide the pdf files for all of the required reading.

Week	Topic	Core Concepts	Required Reading
1	**An Introduction to Biodiversity**	Unequal distribution of biodiversity; measures of diversity; pre-Darwinian views; Darwin's insight and what he lacked; natural selection; the Modern Synthesis	1. *Darwin, C. (1859) Introduction. In, On the Origin of Species.* 2. *Darwin, C. (1859) Chapter IV – Natural Selection. In, On the Origin of Species.* 3. *Hutchinson, G.E. (1959). Homage to Santa Rosalia or why are there so many kinds of animals? Amer. Nat. 93:145-159.* 4. *Laland, K. et al. (2014). Does evolutionary theory need a rethink? Nature 514:161-164.*
2	**What Is a Species and How Are They Formed?**	Species concepts; speciation mechanisms; reproductive isolation (pre- vs. post-zygotic; intrinsic vs. extrinsic); the speciation continuum; geography of speciation; taxonomy vs. evolutionary biology	1. *Ehrlich, P.R., & Raven P.H. (1969) Differentiation of populations: Gene flow seems to be less important in speciation than the neo-Darwinians thought. Science 165, 1228–1232* 2. *Schluter, D. (2001). Ecology and the origin of species. TREE 16:372-380.* 3. *Coyne, J.A. & Orr, H.A. (2004) Speciation, Chs. 1-2.* 4. *Anderson, S.A. & Weir, J.T. (2022). Divergent ecological adaptation during allopatric speciation. Science 378:1214-1218.*
3	**The Outcomes of Secondary Contact**	Secondary contact as the "moment of truth"; species fusion; competitive and reproductive exclusion; character displacement; reinforcement; species sorting; hybridization; genomic introgression	1. *Brown, W.L. & Wilson, E.O. (1956). Character displacement. Syst. Biol. 5:49-64.* 2. *Servedio, M.R. & Noor, M.A.F. (2003). The role of reinforcement in speciation. Ann. Rev. Ecol. Evol. Syst. 34:339-364.* 3. *Seehausen, O., 2006. African cichlid fish: a model system in adaptive radiation research. Proceedings of the Royal Society B: Biological Sciences, 273(1597), p.1987.* 4. *Anderson, S.A. and Matute, D.R., 2025. Secondary sympatry as a sorting process. Ecology Letters, 28(4), p.e70108.*
4	**How Are Communities Assembled?**	Community definition; niche theory; ecological community assembly; priority effects; forbidden pairs; environmental filtering; null models; neutral theory; modern coexistence theory;	1. *Connor, E.F. & Simberloff, D. (1979). Community assembly: chance or competition? Ecology 60:1132-1140.* 2. *Schoener, T.W. (1974). Resource partitioning in ecological communities. Science 185:27-39.* 3. *Lewin, R. (1983). Santa Rosalia was a goat. Science 221:636-639.*

Week	Topic	Core Concepts	Required Reading
		niche and resource partitioning; beta vs. alpha diversity	4. *Rosindell, J., Hubbell, S.P. & Etienne, R.S. (2011). The unified neutral theory at age ten. TREE 26:340-348.* 5. *Fukami, T. (2015). Historical contingency in community assembly. Ann. Rev. Ecol. Evol. Syst. 46:1-23.*
5	**Evolution on Ecological Timescales**	Natural selection; local adaptation; the Lande-Arnold statistical framework; selection gradients; heritability; phenotypic plasticity; measuring selection in the wild; paradox of stasis	*1. Boag, P.T. and Grant, P.R., 1981. Intense natural selection in a population of Darwin's finches (Geospizinae) in the Galapagos. Science, 214(4516), pp.82-85.* *2. Losos, J.B., Warheitt, K.I. and Schoener, T.W., 1997. Adaptive differentiation following experimental island colonization in Anolis lizards. Nature, 387(6628), pp.70-73.* 3. *Reznick, D.N., Losos, J. & Travis, J. (2019). From low to high gear: a paradigm shift in our understanding of evolution. Ecology Letters 22:233-244.* 4. *Svensson, E.I. (2023). Phenotypic selection in natural populations: what have we learned in 40 years? Evolution 77:1493-1504.*
6	**Evolutionary Radiations**	Ecological theory of adaptive radiation; ecological opportunity; ecological release; key innovations; adaptive divergence; resource partitioning; parallel and convergent evolution; adaptive landscape; non-adaptive radiation	*1. Schluter, D. (1996). Ecological causes of adaptive radiation. Amer. Nat. 148:S40-S64.* *2. Schluter, D. (2000). The Ecology of Adaptive Radiation, Ch. 1.* *3. Stroud, J.T. & Losos, J.B. (2016). Ecological opportunity and adaptive radiation. Ann. Rev. Ecol. Evol. Syst. 47:507-532.* *4. Rundell, R.J. & Price, T.D. (2009). Adaptive radiation, nonadaptive radiation, ecological speciation and nonecological speciation. TREE 24:394-399.* *5. Grant, P.R. & Grant, B.R. (2024). From microcosm to macrocosm: adaptive radiation of Darwin's finches. Evol. J. Linn. Soc. 3:kzae006.*
7	**Biological Patterns Across Space**	Major biogeographic patterns; historical explanations; latitudinal diversity gradient; elevational gradients; endemism; dispersal limitation; vicariance; range expansions and contractions; geology and biodiversity (Wallace's Line, plate tectonics)	*1. Wallace, A.R. (1876). The Geographical Distribution of Animals [selected reading TBD].* 2. *MacArthur, R.H. (1965). Patterns of species diversity. Biol. Rev. 40:510-533.* 3. *Pianka, E.R. (1966). Latitudinal gradients in species diversity. Amer. Nat. 100:33-46.* 4. *Janzen, D.H. (1967). Why mountain passes are higher in the tropics. Amer. Nat. 101:233-249.*

Week	Topic	Core Concepts	Required Reading
8	**Island Biogeography Theory**	Dispersal; colonization; extinction; island syndrome; equilibrium theory; species-area relationships; isolation effects; disharmonious faunas; relaxation curves; nested subsets	1. *MacArthur, R.H. & Wilson, E.O. (1963). An equilibrium theory of insular zoogeography. Evolution 17:373-387.* 2. *Losos, J.B. and Schluter, D., 2000. Analysis of an evolutionary species–area relationship. Nature, 408(6814), pp.847-850.* 3. *Helmus, M.R., Mahler, D.L. & Losos, J.B. (2014). Island biogeography of the Anthropocene. Nature 513:543-546.* 4. *Valente, L. et al. (2020). A simple dynamic model explains the diversity of island birds worldwide. Nature 579:92-96.*
9	**Phylogenetics and Macroevolution**	Phylogenetic reconstruction; measuring phylogenetic diversity; trait evolution; over- and underdispersion; phylogenetic signal; comparative methods; diversification rates; key innovations and diversity	1. *Felsenstein, J., 1985. Phylogenies and the comparative method. The American Naturalist, 125(1), pp.1-15.* 2. *Losos, J.B., 2011. Seeing the Forest for the Trees: The Limitations of Phylogenies in Comparative Biology. The American Naturalist, 177(6), pp.709-727.* 3. *Rabosky, D.L., 2017. Phylogenetic tests for evolutionary innovation: the problematic link between key innovations and exceptional diversification. Philosophical Transactions of the Royal Society B: Biological Sciences, 372(1735), p.20160417.*
10	**The Ecology and Evolution of Behavior**	Tinbergen's four questions; adaptive behavior; phenotypic plasticity; behavioral syndromes; ecological and evolutionary consequences of behavior; sociality; altruism; inclusive fitness; kin selection	1. *Tinbergen, N. (1963). On aims and methods of ethology. Z. Tierpsychol. 20:410-433.* 2. *Hamilton, W. D. (1963). The evolution of altruistic behavior. American Naturalist 97:354–356.* 3. *Gould, S. J., & Lewontin, R. C. (1979). The spandrels of San Marco and the Panglossian paradigm: a critique of the adaptationist programme. Proceedings of the Royal Society B 205:581–598*
11	**Eco-Evolutionary Dynamics**	Historical development of eco-evolutionary dynamics; ecological feedbacks from rapid evolution; contemporary evolution; community and ecosystem consequences of evolution; anthropogenic selective pressures; predictive frameworks	1. *Travis, J. et al. (2014). Do eco-evo feedbacks help us understand nature? Advances in Ecological Research 50:1-40.* 2. *Pelletier, F., Garant, D. & Hendry, A.P. (2009). Eco-evolutionary dynamics. Phil. Trans. R. Soc. B 364:1483-1489.* 3. *Bassar, R.D. et al. (2021). Towards a more precise and accurate view of eco-evolution. Ecology Letters 24:623-625.*

Week	Topic	Core Concepts	Required Reading
12	**Final oral exam**		*Students will be tested orally in Qualifying Exam style by the course instructors*